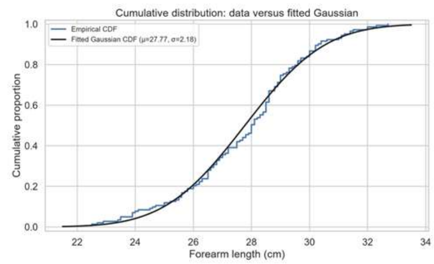
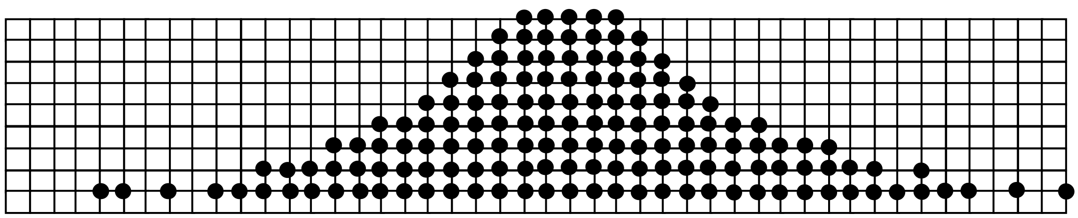
Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



Forearm length of ~150 students

- Last time
  - Normal distribution
- Today
  - More on normal distribution (why?)



Plot of pulse rate of ~150 students

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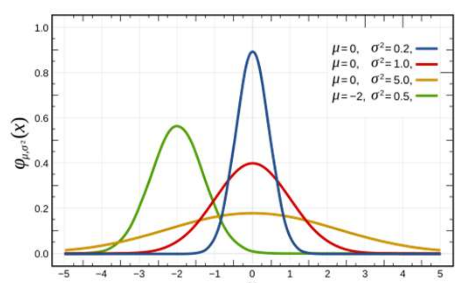
Normal Distribution

- A continuous random variable with parameters mean μ and variance σ^2
  - This pdf is symmetric about the mean μ and has the shape of the "bell curve".
  - If μ=0 and σ=1, we have the standard normal distribution
- We take the positive square root of the variance, conventionally called the standard deviation



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

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Pulse-rate of 50 students

Can calculate statistic such as sample mean [79.1] and sample standard deviation [7.6] but we can do more!

Pulse-rate (beats per minute) of 50 students									
89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

Let's plot these numbers

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sample mean (79.1) and sample standard deviation (7.6)

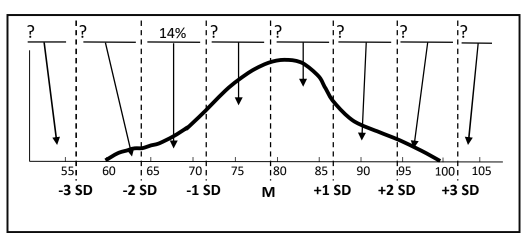
Fill in the "?" mark values!

Example

Percentages: 0, 2, 14, 34, 32, 16, 2, 0

- Consider the 50 students
  - Data has been sorted
  - Underlined 7 data are not measured but computed for reference
- We can bin data based on standard deviations
  - 7 observations (out of 50) between -1 and -2
- … and classify data as, e.g., low pulse rate

56.3	62	63.9	64	65	66	68	70	71	71
71.5	72	72	73	74	74	75	75	76	77
77	77	78	78	78	79	79	79	79.1	80
80	80	80	81	81	81	81	82	82	82
83	83	85	85	86	86.7	87	87	88	89
90	90	92	94	94.3	96	101.9			



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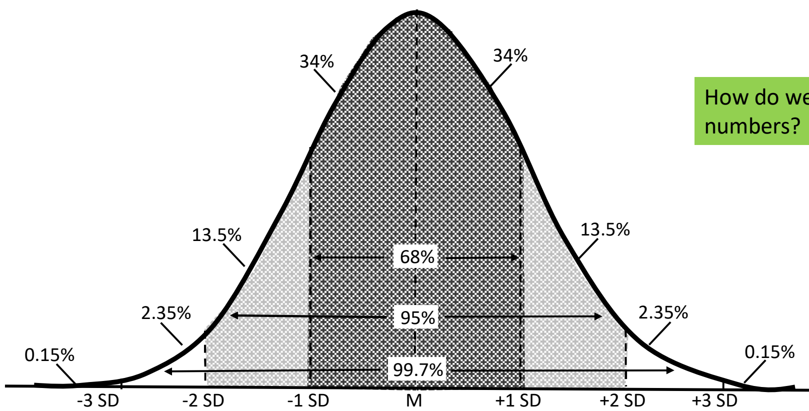
Normal Distribution



How do we get these numbers?

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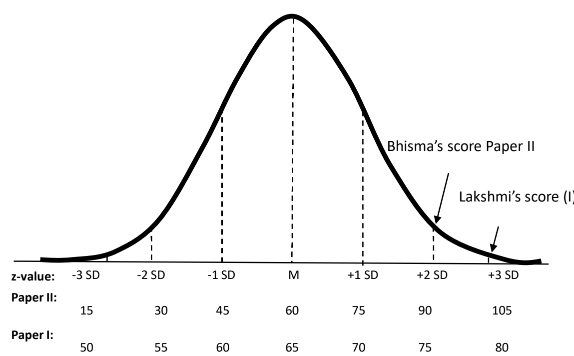
Mahalanobis distance

- Suppose two students Lakshmi and Bhisma, in JEE Mains Papers I & II, got 80% marks and 90% marks respectively
  - Which student should be ranked higher?
- The generalization of $\frac{(x-\mu)}{\sigma}$ in higher dimensions is the Mahalanobis distance
- The z-score of x is the number of standard deviations x is from the mean



29 June is celebrated as **National Statistics Day** (birth anniversary of P.C. Mahalanobis)

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Properties

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, what is pdf of Y?

Proof on board

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, then $Y \sim N(a\mu + b, a^2\sigma^2)$

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Properties

- Z is a standard, or unit normal distribution
- All probability statements of X can be made in terms of probabilities of Z
- Therefore, we study $F_Z(z)$

If $X \sim N(\mu, \sigma^2)$ and $Z = \dfrac{X - \mu}{\sigma}$, then
$Z \sim N(0, 1)$

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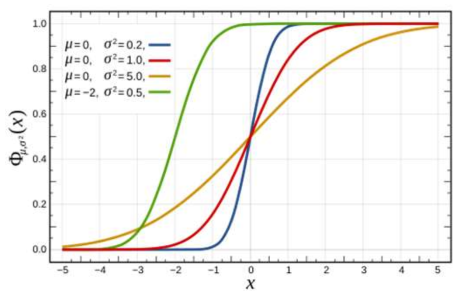
Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- $Z = \frac{(X-\mu)}{\sigma}$ is a standard, or unit normal distribution
- Cumulative distribution function for the standard normal
  - Cannot get numbers like 68-95-99 without numerical integration
  - Closely related to the “erf” function



$$\Phi(x) = F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{x^2}{2}} dx$$

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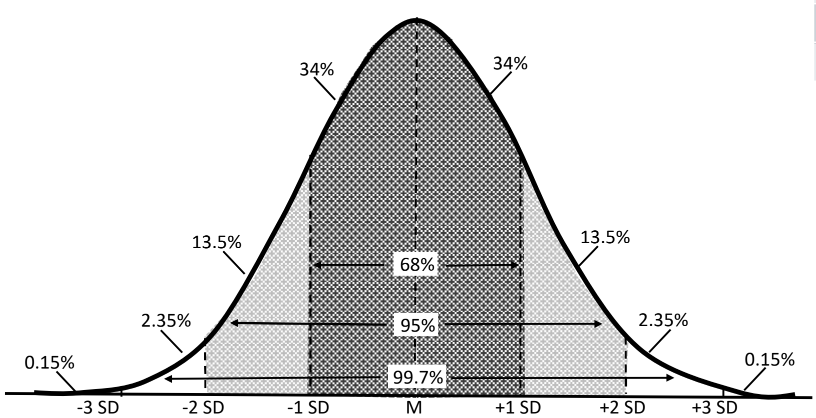
Normal Distribution

n	Φ(*n*)-Φ(-*n*)
1	68.2%
2	95.4%
3	99.7%
4	99.99366%
5	99.9999%
6	99.9999998%



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Example

- Find the probability X < 180 when μ=174 and σ=6.57
  - In calculations involving a continuous random variable, we usually do not have to worry about whether interval endpoints are included or excluded
  - normcdf(180, 174, 6.57)
    - ans=0.8194

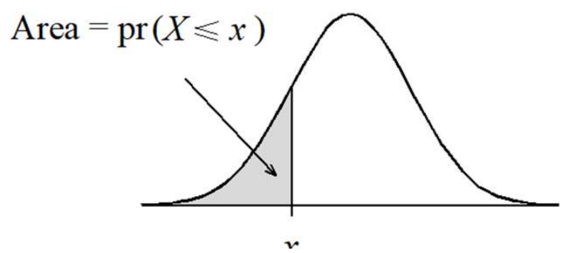

TABLE A1 *Standard Normal Distribution Function:* $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-y^2/2}\, dy$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621

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Basic method for obtaining probabilities

- Sketch a Normal curve, marking the mean and other values of interest.
- Shade the area under the curve that gives the desired probability.
- Devise a way of getting the desired area from lower-tail areas.
- Obtain component lower-tail probabilities from a computer program (or in the exam, provided tables)

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035

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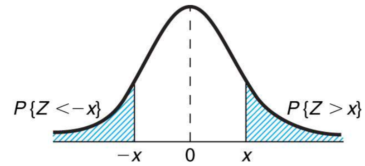
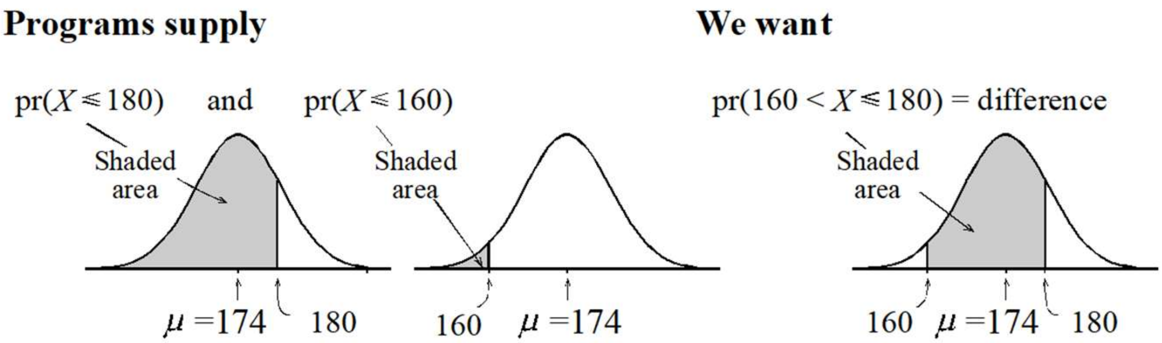
Example



- Find the probability 160 < X < 180 when μ=174 and σ=6.57



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Example

- Find $\Pr(z \le 1.1357)$
- Correct the z-value to two decimal places z=1.14
  1. Look down the first column “z” to find 1.1. This give you the row.
  2. Look across with the second decimal place to find the required cell

z	0	1	2	3	4	5	6	7	8	9
1.0	.841	.844	.846	.848	.851	.853	.855	.858	.860	.862
1.1	.864	.867	.869	.871	.873	.875	.877	.879	.881	.883
1.2	.885	.887	.889	.891	.893	.894	.896	.898	.900	.901
1.3	.903	.905	.907	.908	.910	.911	.913	.915	.916	.918
1.4	.919	.921	.922	.924	.925	.926	.928	.929	.931	.932

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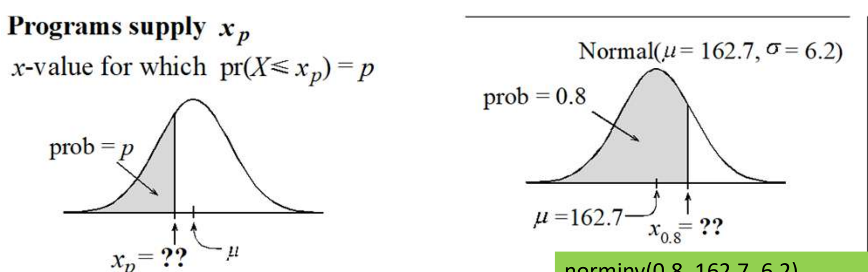
Inverse probability: p-Quantile

- Find the woman whose height is at the 80th percentile if the mean is 162.7 and standard deviation is 6.2



norminv(0.8, 162.7, 6.2)
Ans = 167.92

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Inverse Probability

- Other than the sample mean and the sample variance, another way to summarize is the percentile
  - Toy example (each data is associated with a percentile)
    - Marks: 100, 90, 75, 70, 60
    - Percentile: 100, 80, 60,40,20
- The sample $100p$ percentile $(0 \le p \le 1)$ is defined as the data value y such that $100p\%$ of the data have a value less than or equal to y, and $100(1-p)$ % of the data have a larger value
  - For a data set with n sample points, the sample $100p$ percentile is that value such that at least np of the values are less than or equal to it.
  - And at least $n(1-p)$ of the values are greater than or equal to it

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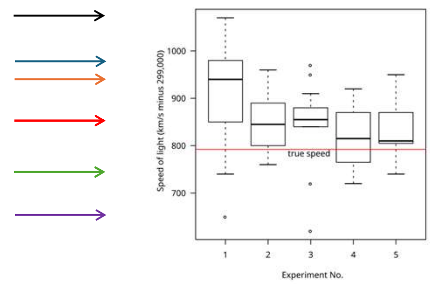
Box Plot

- The sample 25 percentile = first quartile, the sample 50 percentile = second quartile, the sample 75 percentile = third quartile
  - Length of the box is IQR: Interquartile range

- Example: 42 values
  - Low is 57, High is 70
  - First quartile: 0.25*42=10.5
  - Median
  - Third: 0.75*42=31.5



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